

Linearized Per-Source Residual Decomposition

1 Notation

A decoder-only transformer $\mathcal{M}$ with L layers, H heads, model dimension d , feed-forward dimension d_{ff} , head dimension $d_h = d/H$, and input sequence of N tokens.

$e_p \in \mathbb{R}^d$	token embedding at position p
$x_p^{(\ell)} \in \mathbb{R}^d$	residual stream at layer ℓ , position p (real forward pass)
$A^{(\ell,h)} \in [0, 1]^{N \times N}$	attention matrix of head h , layer ℓ (frozen)
$W_{\text{OV}}^{(\ell,h)} \in \mathbb{R}^{d \times d}$	OV circuit: $W_O^{(\ell,h)} W_V^{(\ell,h)}$
$r_p^{(\ell,k)} \in \mathbb{R}_{>0}$	inverse-RMS scale, layer ℓ , position p ; $k=1$ before attention, $k=2$ before MLP (frozen)
$\gamma^{(\ell,k)} \in \mathbb{R}^d$	RMSNorm learned gain, layer ℓ , sub-block k
$g_p^{(\ell)} \in \mathbb{R}^{d_{\text{ff}}}$	frozen gate: $\text{SiLU}(W_{\text{gate}}^{(\ell)} x_p^{(\ell)})$
$c_{p,s}^{(\ell)} \in \mathbb{R}^d$	per-source contribution of token s to position p at layer ℓ

The contributions satisfy $x_p^{(\ell)} = \sum_{s=0}^{N-1} c_{p,s}^{(\ell)}$ and $c_{p,s}^{(\ell)} = 0$ for $s > p$.

2 Algorithm

Initialization ($\ell = 0$).

$$c_{p,s}^{(0)} = \begin{cases} e_p & s = p \\ 0 & s \neq p \end{cases} \quad (1)$$

Layer recurrence (for $\ell = 0, \dots, L-1$). Normalize a contribution vector via frozen RMSNorm:

$$\hat{v}_p^{(\ell,k)} := \gamma^{(\ell,k)} \odot (r_p^{(\ell,k)} \cdot v). \quad (2)$$

Same scale $r_p^{(\ell,k)}$ as applied to the real activation $x_p^{(\ell)}$.

Attention step ($k = 1$):

$$c_{p,s}^{\text{mid},(\ell)} = c_{p,s}^{(\ell)} + \sum_{h=1}^H \sum_{s'=0}^p A_{p,s'}^{(\ell,h)} \cdot W_{\text{OV}}^{(\ell,h)} \hat{c}_{s',s}^{(\ell,1)}. \quad (3)$$

post attention contribution = residual stream + sum over all heads of attention contribution across position from same original source

MLP step ($k = 2$):

$$c_{p,s}^{(\ell+1)} = c_{p,s}^{\text{mid},(\ell)} + W_{\text{down}}^{(\ell)} \left(g_p^{(\ell)} \odot W_{\text{up}}^{(\ell)} c_{p,s}^{\text{mid},(\ell,2)} \right). \quad (4)$$

Residual plus up-project, scale by frozen gate $g_p^{(\ell)}$, down-project.

Output. Tensor $c \in \mathbb{R}^{(L+1) \times N \times N \times d}$. The visualization shows $\|c_{p,s}^{(\ell)}\|_q$ for $q \in \{0, 1, 2, \infty\}$.